Derivation of an Effective Convective Heat Transfer Coefficient for Lumped Thermal Modeling

Objective

The goal is to determine a single, effective convective heat transfer coefficient (h_{eff}) from discrete, time-dependent heat transfer data recorded across five distinct surfaces. This value will be utilized in a lumped capacitance thermal model, which requires a single time-invariant and spatially-averaged heat transfer coefficient.

Methodology and Justification

1. Time-Averaging via Trapezoidal Integration

The governing energy balance equation for convective heat transfer in a lumped system is:

$$\rho V c_p \frac{dT}{dt} = -\sum_{i=1}^{n} h_i(t) A_i (T - T_\infty) \tag{1}$$

Because the model demands a single constant h over the entire simulation duration (t_0 to $t_{\max}$), a time-averaged value for each surface i must first be established. Using trapezoidal numerical integration normalized by total time:

$$\bar{h}_i = \frac{1}{t_{\max} - t_0} \int_{t_0}^{t_{\max}} h_i(t)\, dt \tag{2}$$

Applying this to the simulated discrete data yields the following steady-state equivalent values:

- Surface 1 (h_1): 107.22 W/(m$^2 \cdot$ K)
- Surface 2 ($h_{1.1}$): 426.60 W/(m$^2 \cdot$ K)
- Surface 3 ($h_{1.2}$): 114.71 W/(m$^2 \cdot$ K)
- Surface 4 ($h_{1.3}$): 116.13 W/(m$^2 \cdot$ K)
- Surface 5 ($h_{1.4}$): 113.94 W/(m$^2 \cdot$ K)

2. Spatial-Averaging via Area-Weighting

Because the five boundaries vary in physical size, an unweighted arithmetic mean would incorrectly skew the global boundary condition. The individual time-averaged coefficients must be proportionally scaled by their respective surface areas (A_i):

$$h_{\text{eff}} = \frac{\sum_{i=1}^{5} \bar{h}_i A_i}{\sum_{i=1}^{5} A_i} \tag{3}$$

The provided surface areas for the corresponding boundaries are:

- $A_1 = 1995$
- $A_2 = 1425$
- $A_3 = 875$
- $A_4 = 1425$
- $A_5 = 875$

The total surface area is $A_{\text{total}} = 1995 + 1425 + 875 + 1425 + 875 = 6595$.

Final Calculation

Substituting the integrated time-averaged values and known surface areas into the area-weighted equation yields:

$$h_{\text{eff}} = \frac{(107.22 \times 1995) + (426.60 \times 1425) + (114.71 \times 875) + (116.13 \times 1425) + (113.94 \times 875)}{6595} \tag{4}$$

$$h_{\text{eff}} = \frac{213903.9 + 607905.0 + 100371.25 + 165485.25 + 99697.5}{6595} \tag{5}$$

$$h_{\text{eff}} = \frac{1187362.9}{6595} = 180.04 \text{ W}/(\text{m}^2 \cdot \text{K}) \tag{6}$$

Conclusion: By sequentially applying trapezoidal time-integration to isolate temporal variation, followed by an area-weighted spatial average to account for asymmetric boundary sizes, 180.04 $\text{W}/(\text{m}^2 \cdot \text{K})$ represents the correct mathematical reduction to govern the lumped thermal model.